PRECALCULUS · UNIT 2: FUNCTIONS

Streaming Service Analyst

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students analyze a streaming service. Pricing plans become piecewise functions, campaigns become transformations, tax and currency conversions become compositions, and subscriber growth uses exponential and logarithmic functions.

Inverses answer "when?" questions about the business.

Standards Addressed

- F-IF.B.4–5 Interpret functions; domain and range
- F-BF.B.3 Transformations
- F-BF.A.1c, B.4 Composition and inverses
- F-LE.A.4 Logarithms to solve exponential equations

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with streaming plan prices. Part 1 piecewise and compositions.
Day 2	Part 2 transformations and properties.
Day 3	Part 3 exponential and log models; briefing.

Materials

- Student handout and Business Data Cards
- Graphing calculators or Desmos
- Slide or poster template

How the Three Versions Differ

Level	What changes
CP2	Piecewise cost, basic transformations, and exponential predictions with tables.
CP1	Compositions, inverses, transformed models, and logarithms to solve for time.
Honors	Average cost from a piecewise model, inverse domains, continuous growth, doubling time, half-life, and strategy comparison.

Watch For (Common Misconceptions)

- Shifting the wrong direction for f(x − h).
- Composing in the wrong order.
- Treating 8% growth as adding 0.08 instead of multiplying by 1.08.

Extension Ideas

- Use real subscriber data from a news article.
- Graph all models in Desmos.
- Add a logistic model with a cap.

Answer Key – CP2

Part 1: Plans as Functions

#	Task	Answer
1	Write the cost C(g) for g GB if $g \le 10$ and if $g > 10$.	$C(g) = 5$ for $0 \le g \le 10$; $C(g) = 5 + 2(g - 10)$ for $g > 10$
2	Find C(6), C(10), and C(14).	\$5, \$5, \$13
3	What is a sensible domain and range for C?	Domain $g \ge 0$; range $C \ge 5$
4	Is C a function? Explain.	Yes; each amount of data has exactly one cost.

Part 2: Transformations

#	Task	Answer
5	Describe how $y = (x - 3)^2 + 2$ is related to $y = x^2$.	Shift right 3 and up 2
6	Describe $y = -2x^2$.	Reflect over the x-axis and stretch vertically by 2
7	Write the equation of $y = \|x\|$ shifted left 4 and down 1.	$y = \|x + 4\| - 1$
8	Basic's price rises \$1 per month. How does the graph of C change?	It shifts up 1.

Part 3: Growth

#	Task	Answer
9	Find S(6) and S(12). Round to the nearest subscriber.	≈ 31,737 and ≈ 50,363
10	Use a table to find the first month the subscribers reach 100,000.	$t = 21$ (≈ 100,677)
11	Is this growth linear or exponential? How do you know?	Exponential; it grows by the same percent (8%) each month.

Reflection

#	Prompt	Look for
R1	How does a piecewise function model a real price plan?	Different rules apply on different intervals of data use.
R2	Why does exponential growth start slow and then speed up?	It adds a percent of a growing amount.

Answer Key – CP1

Part 1: Pricing Functions

#	Task	Answer
1	Write the Basic plan cost as a piecewise function and graph it.	$C(g) = 5$ $(0 \le g \le 10)$; $5 + 2(g - 10)$ $(g > 10)$
2	Write the cost with tax, $T(C(g))$, for $g = 14$.	$T(13) = 1.0625 \times 13 \approx \$13.81$
3	Write $E(T(x))$ as a single function and find the euro price of the \$11 Premium plan.	$E(T(x)) = 0.92(1.0625x) = 0.9775x$; ≈ €10.75
4	Find the inverse of $E(T(x))$. What does it convert?	$x = \frac{y}{0.9775}$; it converts euros (with tax) back to dollars before tax.

Part 2: Transform the Data

#	Task	Answer
5	A marketing campaign adds 5,000 subscribers at launch. Write the new function and describe the transformation.	$S(t) + 5{,}000$ (shift up) — or $20{,}000(1.08)^t + 5{,}000$
6	The campaign starts 3 months late. Write $S(t - 3)$ and describe the shift.	$20{,}000(1.08)^{t-3}$; right 3
7	Is $S(t)$ even, odd, or neither? Is it one-to-one?	Neither; yes, it is one-to-one (increasing).
8	Find the domain and range of $S(t)$ for $t \ge 0$.	Domain $t \ge 0$; range $S \ge 20{,}000$

Part 3: Logs & Goals

#	Task	Answer
9	Solve $20{,}000(1.08)^t = 100{,}000$ using logarithms.	$t = \frac{\ln 5}{\ln 1.08} \approx 20.9$ months
10	Old app users: $U(t) = 50{,}000(0.95)^t$. When do they fall below 25,000?	$t > \frac{\ln 0.5}{\ln 0.95} \approx 13.5 \rightarrow$ month 14
11	When will new subscribers be twice the old app users?	$20{,}000(1.08)^t = 2 \times 50{,}000(0.95)^t \rightarrow (\frac{1.08}{0.95})^t = 5 \rightarrow t \approx 12.5$ months

Reflection

#	Prompt	Look for
R1	What does the composition $E(T(x))$ mean in words?	Add tax first, then convert the taxed price to euros.

#	Prompt	Look for
R2	Why do we use logarithms to solve for time in an exponential model?	The variable is in the exponent; logarithms bring it down.

Answer Key – Honors

Part 1: Model Building

#	Task	Answer
1	A Family plan is $17 for up to 4 users and $3.50 per extra user. Write F(u) as a piecewise function and find the average cost per user for 6 users.	F(u) = 17 (1 ≤ u ≤ 4); 17 + 3.5(u − 4) (u > 4). F(6) = 24 → $4 per user
2	For how many users is the Family plan's average cost per user less than the $5 Basic price?	For u ≤ 4: $\frac{17}{u} < 5 \rightarrow u > 3.4$, so u = 4 ($4.25). For u > 4 the average is $3.5 + \frac{3}{u} < 5$. So 4 or more users.
3	Find the inverse of C(g) = 5 + 2(g − 10) for g > 10 and state its domain.	$C^{-1}(c) = \frac{c-5}{2} + 10$ for c > 5
4	Explain why the whole Basic-plan function C(g) (including 0 ≤ g ≤ 10) has no inverse.	It is constant on [0, 10]; many inputs give $5, so it is not one-to-one.

Part 2: Growth Models

#	Task	Answer
5	Rewrite S(t) as a continuous model $20{,}000e^{kt}$. Find k.	k = ln 1.08 ≈ 0.0770
6	Find the doubling time of S and the half-life of U.	$\frac{\ln 2}{\ln 1.08}$ ≈ 9.0 months; $\frac{\ln 0.5}{\ln 0.95}$ ≈ 13.5 months
7	Total users are S(t) + U(t). Find the month when total users are lowest (use a table).	t = 0: 70,000; t = 3: ≈ 68,063; t = 4: ≈ 67,935; t = 5: ≈ 68,076. The total is lowest at month 4 (about 67,935 users).
8	Find the inverse of S(t). What question does $S^{-1}(n)$ answer?	$S^{-1}(n) = \frac{\ln(n \div 20{,}000)}{\ln 1.08}$; the month when there are n subscribers

Part 3: Strategy

#	Task	Answer
9	Strategy A: 10% monthly growth from 20,000. Strategy B: linear growth of 3,000 per month from 30,000. When does A pass B?	$20{,}000(1.1)^t = 30{,}000 + 3{,}000t \rightarrow t \approx 13.0$ months (solve with a table or graph)
10	Which strategy reaches 100,000 first, and when?	A: $\frac{\ln 5}{\ln 1.1}$ ≈ 16.9 months; B: $\frac{70{,}000}{3{,}000}$ ≈ 23.3 months. A is first.
11	Write a briefing recommending a strategy, with graphs and at least two function concepts.	Answers vary.

Reflection

#	Prompt	Look for
R1	Why must a function be one-to-one to have an inverse function?	Otherwise one output would come from several inputs, and the inverse would not be a function.
R2	How do the doubling time and half-life summarize growth and decay?	They give the constant time for the quantity to double or halve, independent of the starting amount.